

CS-223: Operating Systems

Couse Teacher: Dr. Ghazanfar Farooq

Objectives:

An operating system is a crucial part of a computer system and is the first piece of software to run on a computer when it is booted. This course is aimed at providing the basic knowledge about essential concepts and issues that lie behind operating systems and their design. The main goals of the course are to develop an understanding of the function, the structure, and operation of modern operating systems, to understand the interaction between (system) software and hardware, as well as between the operating system and applications and to get some practical experience in operating system development and the associated difficulties and techniques.

Lecture Schedule:

Week	Lecture
1, 2	Introduction to Operating System: Types and History of Operating Systems.
3, 4, 5	Process Management: Process Vs. Thread, Process/Thread Scheduling. Process models, five state and seven state models CPU Scheduling: Scheduling criteria, Scheduling algorithms,
6, 7	Process Synchronization: Concurrency Problem. Software/Hardware Solutions for Critical Section Problem. Semaphores. Use of semaphores for Critical Section Problem.
8,9	Classical Synchronization Problems. Bounded Buffer Problem. Readers/Writers Problem. Dining Philosophers Problem. Monitors. Inter Process Communication. Implementation of PTHREADs
9,10	Deadlocks: Methods for Handling Deadlocks. Deadlock Prevention. Deadlock Avoidance. Deadlock Detection and Recovery.
11, 12	Memory Management: Contiguous Memory Allocation. Fixed/Variable Partitions Allocation Schemes. Paging and Segmentation.
13, 14	Virtual Memory. Demand Paging. Page Replacement Policies. Demand Segmentation.
15, 16	Storage Management: Overview of Mass-Storage Structure, Disk Scheduling, RAID Structure, Directory Structure. File-System Structures. Allocation Methods. Free Space Management. Protection and Security

Evaluation Criteria

Assignments & Quizzes	:	25%
Mid	:	25%
Terminal	:	50%

Recommended Readings:

- Silberschatz et al, Operating System Concepts.2018, 10th edition.
- William Stallings, Operating Systems, 9th Edition 2018.